



Classification



A person arrives at the emergency room with some symptoms that could be attributed to one of three medical conditions. Which condition does the person have?

An online banking service must determine whether a transaction performed on it is fraudulent, on the basis of the user IP address, transaction history, etc.

Based on information such as client location, local traffic, etc. Coupang wants to predict whether a package can be delivered to a client by 4 PM the next day.



Classification

Linear regression assumes a quantitative response Y

If the response is qualitative (categorical), we have a *classification* problem

Predicting a qualitative response is like *classifying* an observation x , assigning it to 1 of K possible categories

Many methods:

Logistic Regression (and its extensions), Linear Discriminant Analysis, Naive Bayes, tree-based methods, random forests, deep learning etc.



Why Not Linear Regression?

Patient at the emergency room, 3 possible diagnosis: stroke, drug overdose, epileptic seizure

We can numerically code the response outcomes, and apply linear regression

$$Y = \begin{cases} 1 & \text{if stroke} \\ 2 & \text{if drug overdose} \\ 3 & \text{if epileptic seizure} \end{cases}$$



Why Not Linear Regression?

The coding implies a relation between the outcomes (ordering)
Changing the coding will change completely the model

$$Y = \begin{cases} 1 & \text{if stroke} \\ 2 & \text{if drug overdose} \\ 3 & \text{if epileptic seizure} \end{cases}$$

$$Y = \begin{cases} 1 & \text{if epileptic seizure} \\ 2 & \text{if drug overdose} \\ 3 & \text{if stroke} \end{cases}$$



Natural Ordering?

If response values have a natural ordering, we can still use linear regression

Mild ($Y = 1$)

Moderate ($Y = 2$)

Severe ($Y = 3$)

Hard to find such a natural ordering in real-world problems



What if $K=2$?

If we have only two values for Y , we can use dummy variables

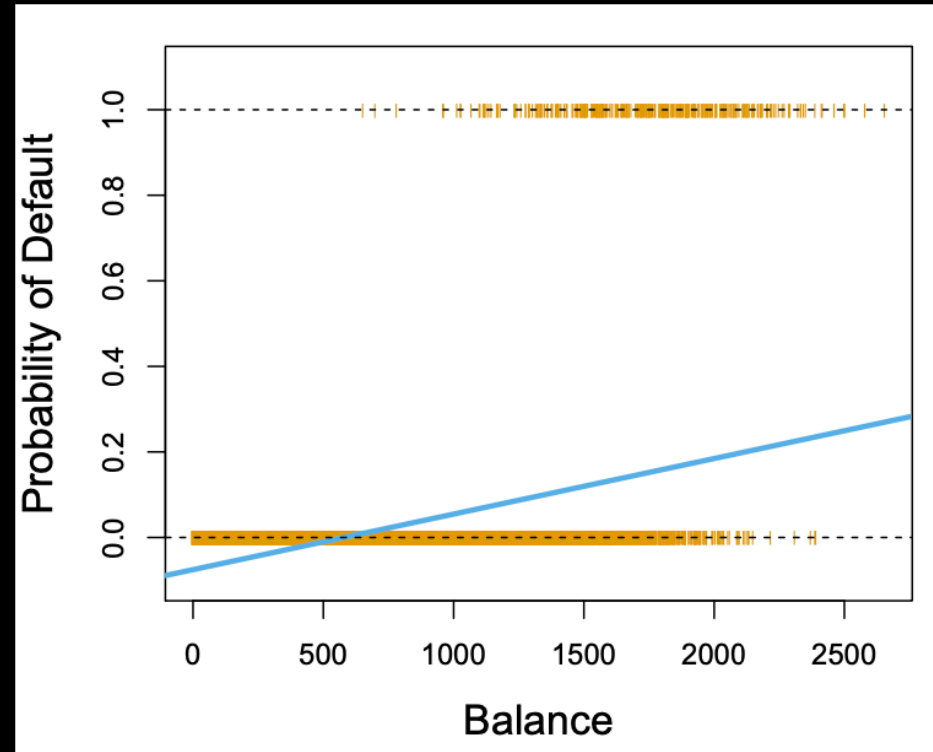
Fit a simple regression, predict drug overdose if $\hat{Y} < 0.5$ and stroke otherwise

Why not?



Problem with Linear Regression as Classifier

Hard to model $p(X)$ using a function that does not output values in $[0, 1]$, to be interpreted as probabilities





What if $K=2$?

If we have only two values for Y , we can use dummy variables

Fit a simple regression, predict drug overdose if $\hat{Y} < 0.5$ and stroke otherwise

OK, but no guarantee that $\hat{Y} \in [0, 1]$

Hard to interpret $\hat{Y}$ as probability

$$Y = \begin{cases} 0 & \text{if stroke} \\ 1 & \text{if drug overdose} \end{cases}$$



Logistic Regression



Example

Default dataset

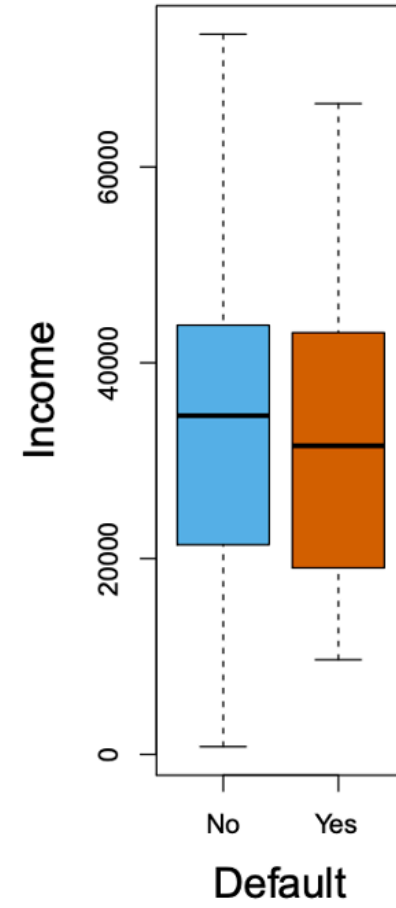
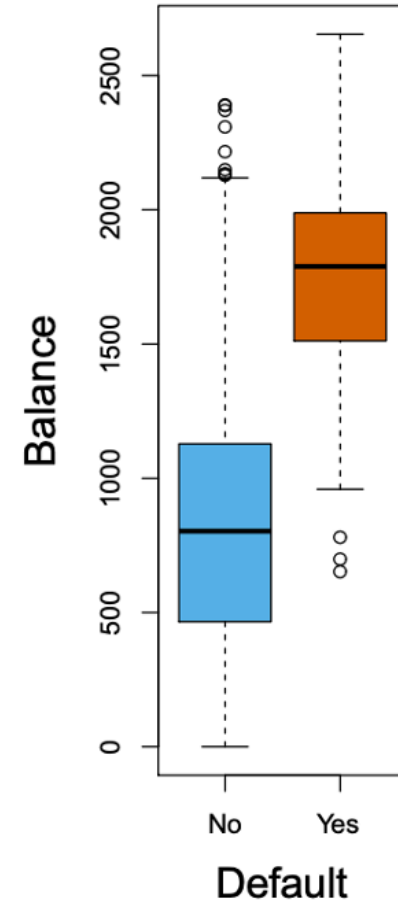
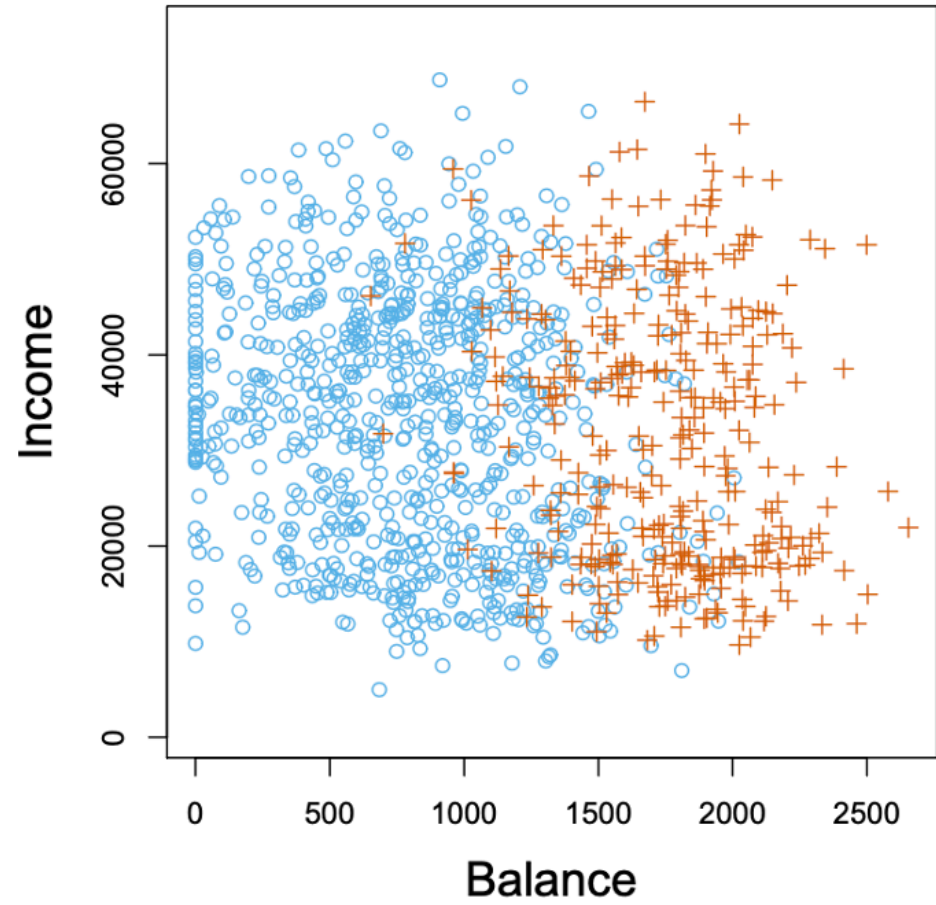
Predictors ($p = 2$)

X_1 Credit Card balance of customer

X_2 Income of customer

Response

Y : will the customer default on credit card debt? YES/NO ($K = 2$)





Logistic Regression (for $p = 1$)

Models the probability that Y belongs to a particular category

We write $Pr(\text{default} = \text{Yes} | \text{balance})$ as $p(\text{balance})$ for simplicity

$$p(\text{balance}) \geq 0.5 \implies \hat{Y} = \text{default}$$

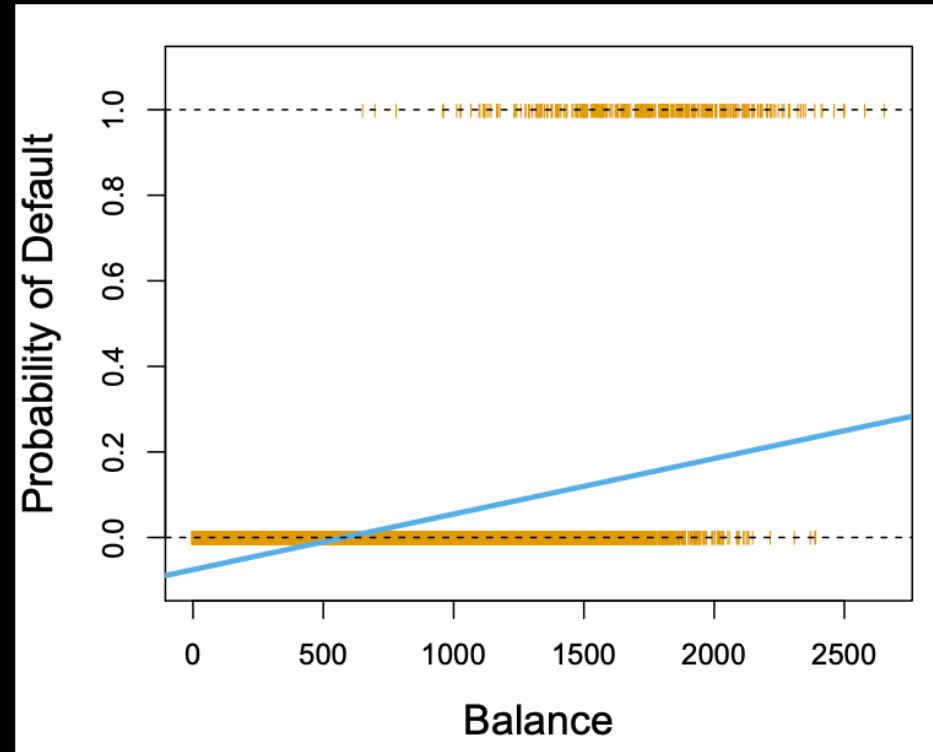
We can also be more conservative:

$$p(\text{balance}) \geq 0.35 \implies \hat{Y} = \text{default}$$



Problem with Linear Regression

We need to model $p(X)$ using a function that outputs values in $[0, 1]$, to be interpreted as probabilities



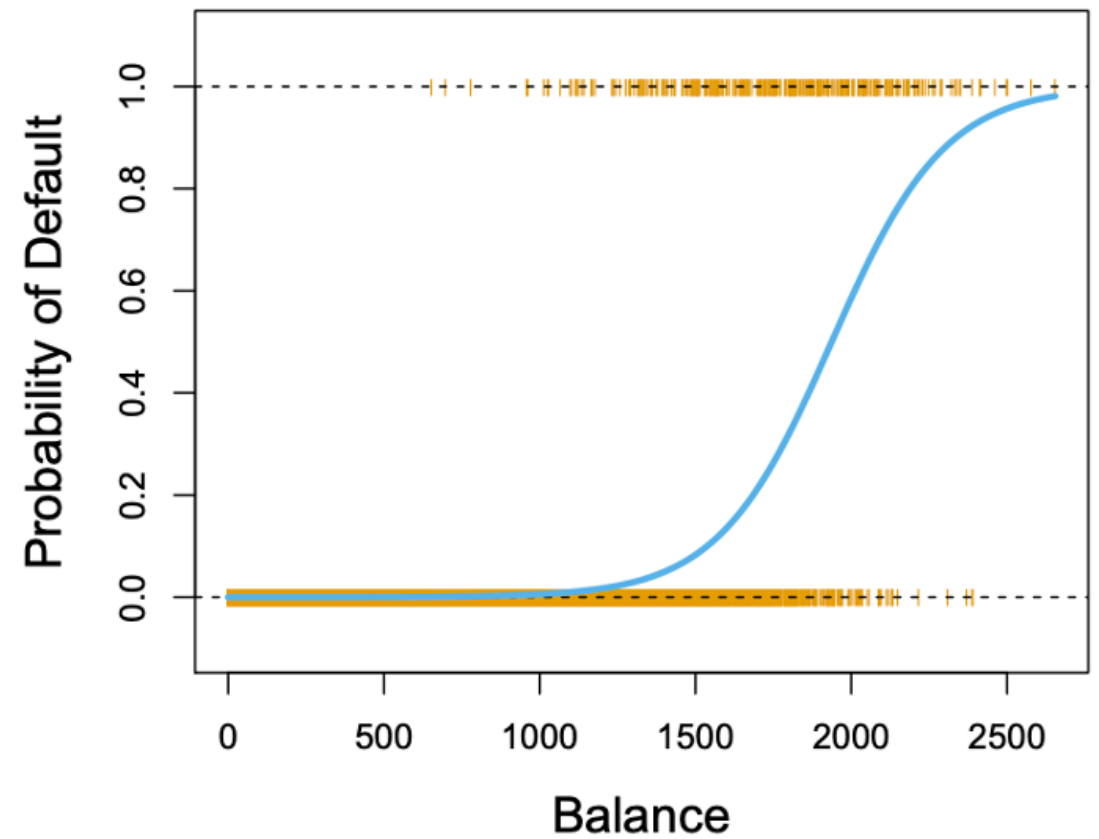
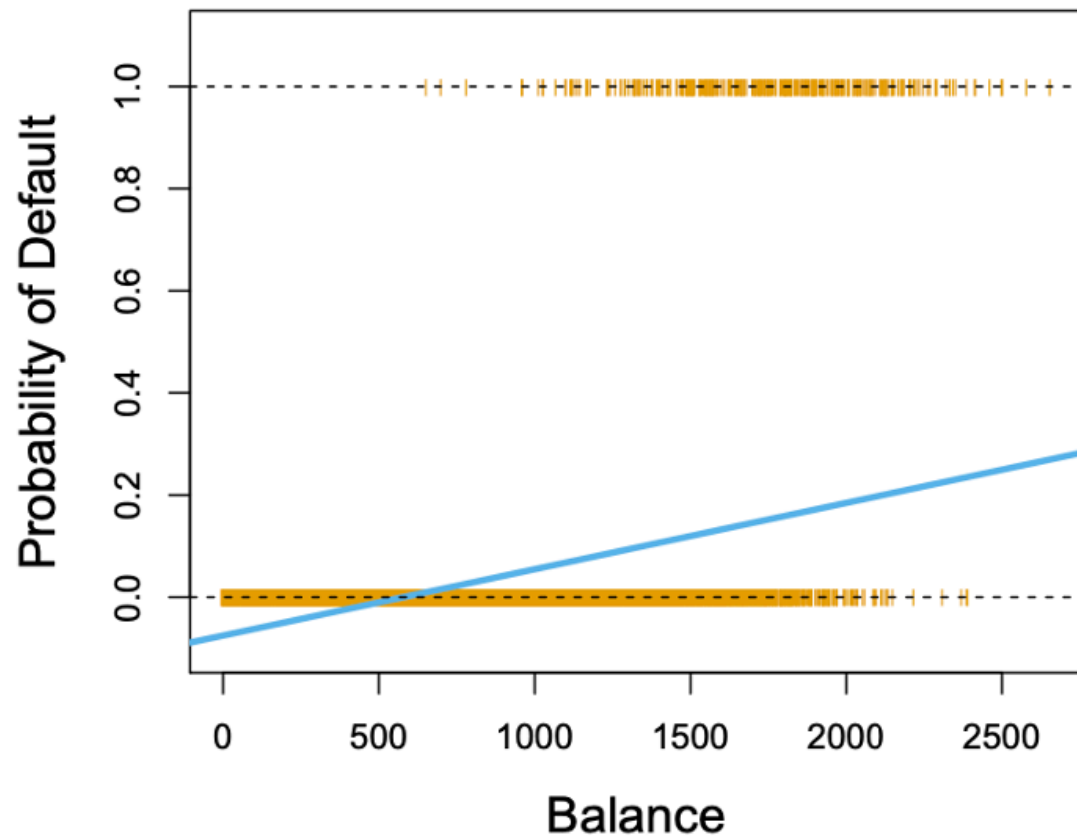


Logistic Regression

Model $p(X)$ using the logistic function

We fit the model using the maximum likelihood method (see later)

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$





Let's Dig into the Logistic Function

$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$ can be rewritten as:

$$\frac{p(X)}{1-p(X)} = e^{\beta_0 + \beta_1 X} \text{ (the odds)}$$

$$\log \left(\frac{p(X)}{1-p(X)} \right) = \beta_0 + \beta_1 X \text{ (the log odds)}$$



Odds in Logistic Regression $\frac{p(X)}{1-p(X)} = e^{\beta_0 + \beta_1 X}$

Odds compare how likely X is to happen with how likely X is *not* to happen

Can take any value between 0 and ∞

0 (∞) indicates very low (high) probability of default

If $p(X) = 0.2$, then odds = $\frac{1}{4} = \frac{0.2}{1-0.2}$: on average 1 in 5 people will default

If, $p(X) = 0.9$, then odds = $9 = \frac{0.9}{1-0.9}$: on average 9 in 10 people will default



Odds and Bets

Odds often used in betting (instead of probabilities) because they are more understandable as betting strategies.

MVP WINNER ODDS						
Filter by name						
	Victor Wembanyama	+200	+200	+225	+175	+200
	Shai Gilgeous-Alexander	+350	+350	+350	+450	+350
	Nikola Jokic	+550	+450	+475	+500	+550
	Luka Doncic	+600	+600	+650	+500	+600

What are the implied probabilities of the 3 events?



Odds and Bets

In (American) odds +S means bet \$100 to win S profit.

The implied probabilities are:

Victor Wenbayama: $p_W = \frac{100}{200+100} = 0.333$

(a 33% chance of winning the MVP Award)

The logistic regression odds are: $odds_W = \frac{0.33}{1-0.33} = 0.5$

(for every one time he wins, there are two times he does not)



Fitting the Model $p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$

How do we estimate the coefficients β_0 and β_1 ?

We could still use *least squares*

Instead we use the *maximum likelihood* method, which has better statistical properties



Maximum Likelihood

The idea:

We look for the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ such that the predicted probability $\hat{p}(x_i)$ of default of each individual corresponds as closely as possible to x_i 's observed default status (1 if x_i defaulted, 0 if x_i did not)

The formula:

Choose $\hat{\beta}_0$ and $\hat{\beta}_1$ that maximize the likelihood function:

$$l(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \prod_{i':y_{i'}=0} (1 - p(x_{i'}))$$

$$\text{with } p(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$



Accuracy of the Statistics

Similar to linear regression

A large value of the z-statistic indicates that the coefficient value is significant

	Coefficient	Std. error	z-statistic	p-value
Intercept	−10.6513	0.3612	−29.5	<0.0001
balance	0.0055	0.0002	24.9	<0.0001



Making Predictions

With an estimate of the coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$, we can make a prediction of the probability of default for any balance value

For balance = 1000\$

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X}} = 0.00576 \text{ (0.5 \%)}$$

For balance = 2000\$, 0.586 (> 50%!)

	Coefficient	Std. error	z-statistic	p-value
Intercept	-10.6513	0.3612	-29.5	<0.0001
balance	0.0055	0.0002	24.9	<0.0001



Qualitative Predictors

Like in linear regression, we can use a dummy variable

Student status:

1: is a student

0: it is not a student

	Coefficient	Std. error	<i>z</i> -statistic	<i>p</i> -value
Intercept	−3.5041	0.0707	−49.55	<0.0001
student [Yes]	0.4049	0.1150	3.52	0.0004



Multiple Logistic Regression



Multiple Logistic Regression

Predicting a binary response using multiple predictors

We simply generalize the (simple) logistic regression

We can use “maximum likelihood” to estimate the coefficients

$$\log \frac{p(X)}{1-p(X)} = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

$$p(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}$$



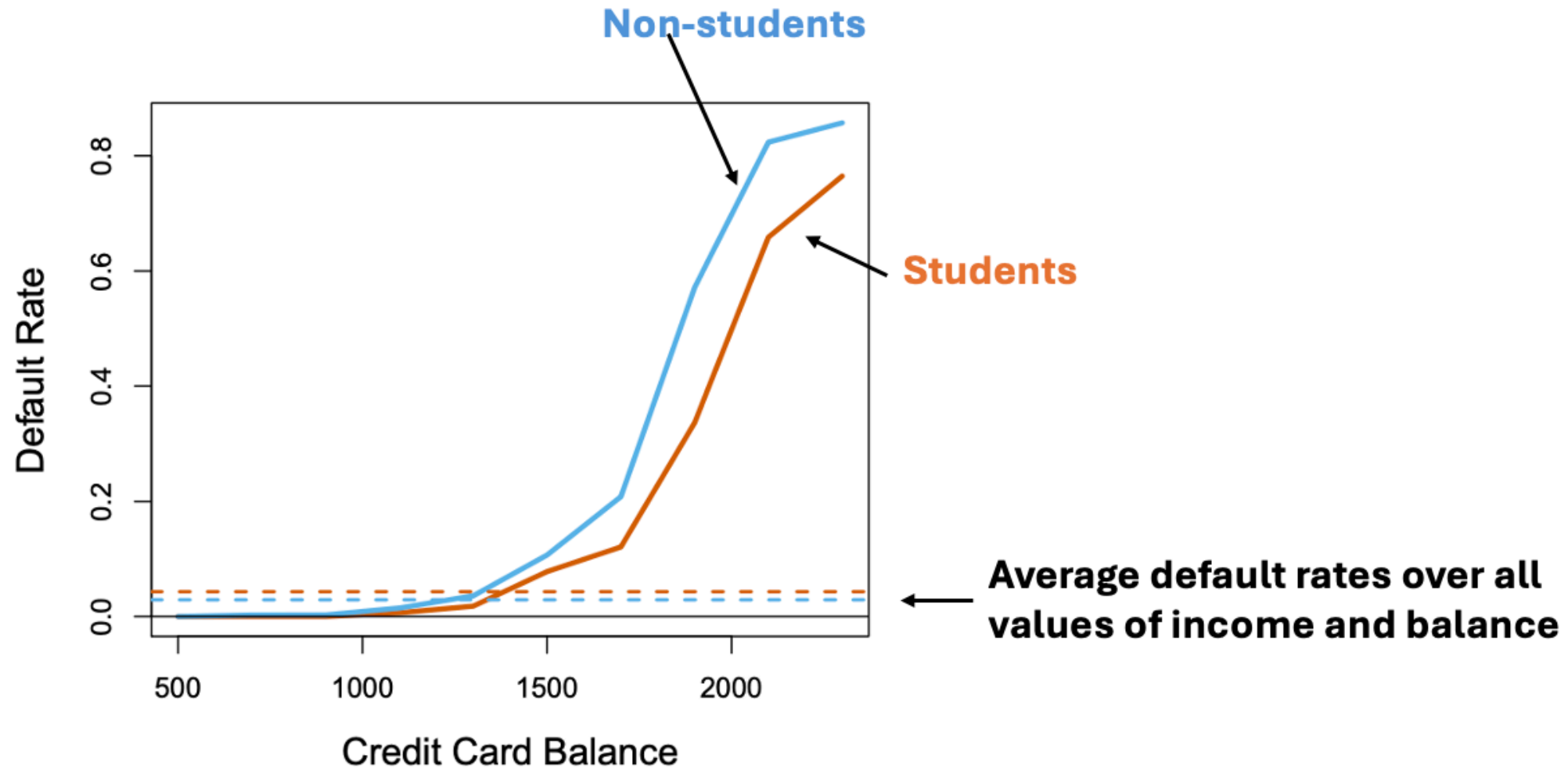
	Coefficient	Std. error	<i>z</i> -statistic	<i>p</i> -value
Intercept	−10.8690	0.4923	−22.08	<0.0001
balance	0.0057	0.0002	24.74	<0.0001
income	0.0030	0.0082	0.37	0.7115
student [Yes]	−0.6468	0.2362	−2.74	0.0062



No Problem?

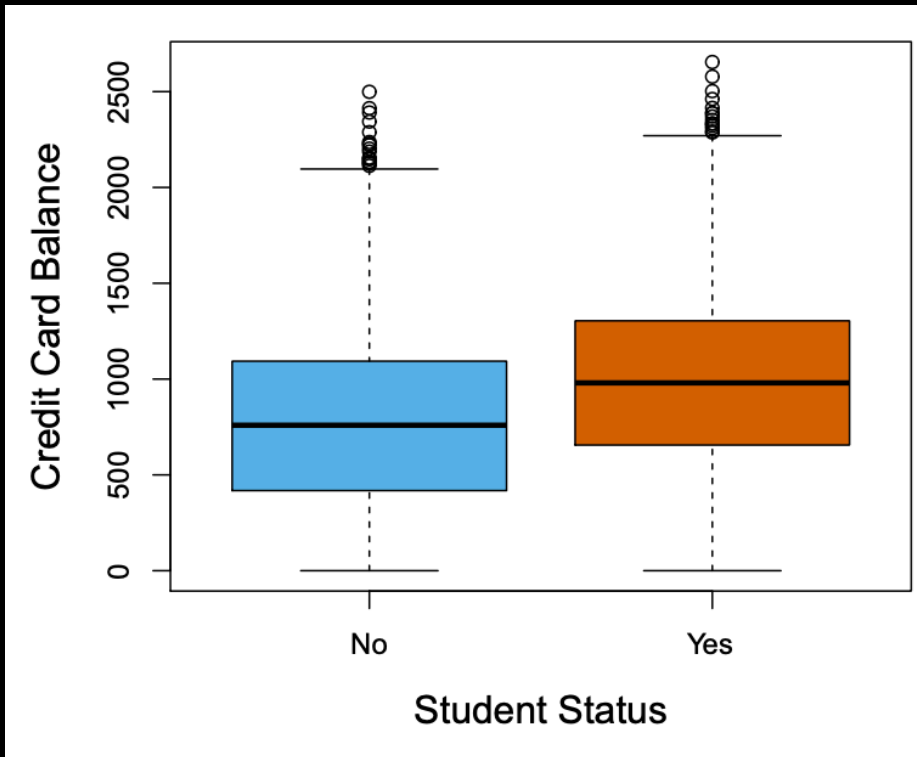
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Balance and student status are highly correlated
Higher balances (= more debt!) are associated with higher default rates





A student is still overall a riskier customer than a non-student for the bank!

	Coefficient	Std. error	z-statistic	p-value
Intercept	−3.5041	0.0707	−49.55	<0.0001
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But fixed the balance and income, a non-student is riskier than a student

$$\hat{p}(X) = \frac{e^{-10.869+0.00574 \times 1,500+0.003 \times 40-0.6468 \times 1}}{1+e^{-10.869+0.00574 \times 1,500+0.003 \times 40-0.6468 \times 1}} = 0.058$$

$$\hat{p}(X) = \frac{e^{-10.869+0.00574 \times 1,500+0.003 \times 40-0.6468 \times 0}}{1+e^{-10.869+0.00574 \times 1,500+0.003 \times 40-0.6468 \times 0}} = 0.105$$

Correlated predictors can be confounding!



Multinomial Logistic Regression



Multinomial Logistic Regression

What if the response has $K > 2$ classes?

Medical condition at emergency room: stroke, overdose, or seizure?

Multinomial Logistic Regression is a straightforward generalization of logistic regression for $K > 2$



Generalization of Logistic Regression

We choose the k -th class as the *baseline*, then for $k = 1, \dots, K - 1$:

$$Pr(Y = k|X = x) = \frac{e^{\beta_{k0} + \beta_{k1}x_1 + \dots + \beta_{kp}x_p}}{1 + \sum_{l=1}^{K-1} e^{\beta_{l0} + \beta_{l1}x_1 + \dots + \beta_{lp}x_p}}$$

For $k = K$

$$Pr(Y = K|X = x) = \frac{1}{1 + \sum_{l=1}^{K-1} e^{\beta_{l0} + \beta_{l1}x_1 + \dots + \beta_{lp}x_p}}$$

Log odds:

$$\log \left(\frac{Pr(Y=k|X=x)}{Pr(Y=K|X=x)} \right) = \beta_{k0} + \beta_{k1}x_1 + \dots + \beta_{kp}x_p$$



Does the Choice of the Baseline Matter?

No, the coefficients change with a different baseline, but the log odds between pairs of classes stay the same

The meaning of the coefficients depends on the baseline



The *softmax* encoding

No baseline is selected, all classes are treated symmetrically:

$$Pr(Y = k|X = x) = \frac{e^{\beta_{k0} + \beta_{k1}x_1 + \dots + \beta_{kp}x_p}}{\sum_{l=1}^K e^{\beta_{l0} + \beta_{l1}x_1 + \dots + \beta_{lp}x_p}}$$

$$\log \left(\frac{Pr(Y=k|X=x)}{Pr(Y=k'|X=x)} \right) = (\beta_{k0} - \beta_{k'0}) + (\beta_{k1}x_1 - \beta_{k'1}) + \dots + (\beta_{kp}x_p - \beta_{k'p})$$